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CSS SELECTORS

More Selector Types

- Descendent selectors
- ID selectors
- Class selectors
- Universal selector

Descendent Selectors

A **descendent selector** targets elements contained in another element.

It's a kind of **contextual selector** (it selects based on relationship to another element).

It's indicated in **a list separated by spaces**

```
ol a {font-weight: bold;}
```

(only the links (**a**) in ordered lists (**ol**) would be bold)

```
h1 em {color: red;}
```

(only emphasized text in h1s would be red)

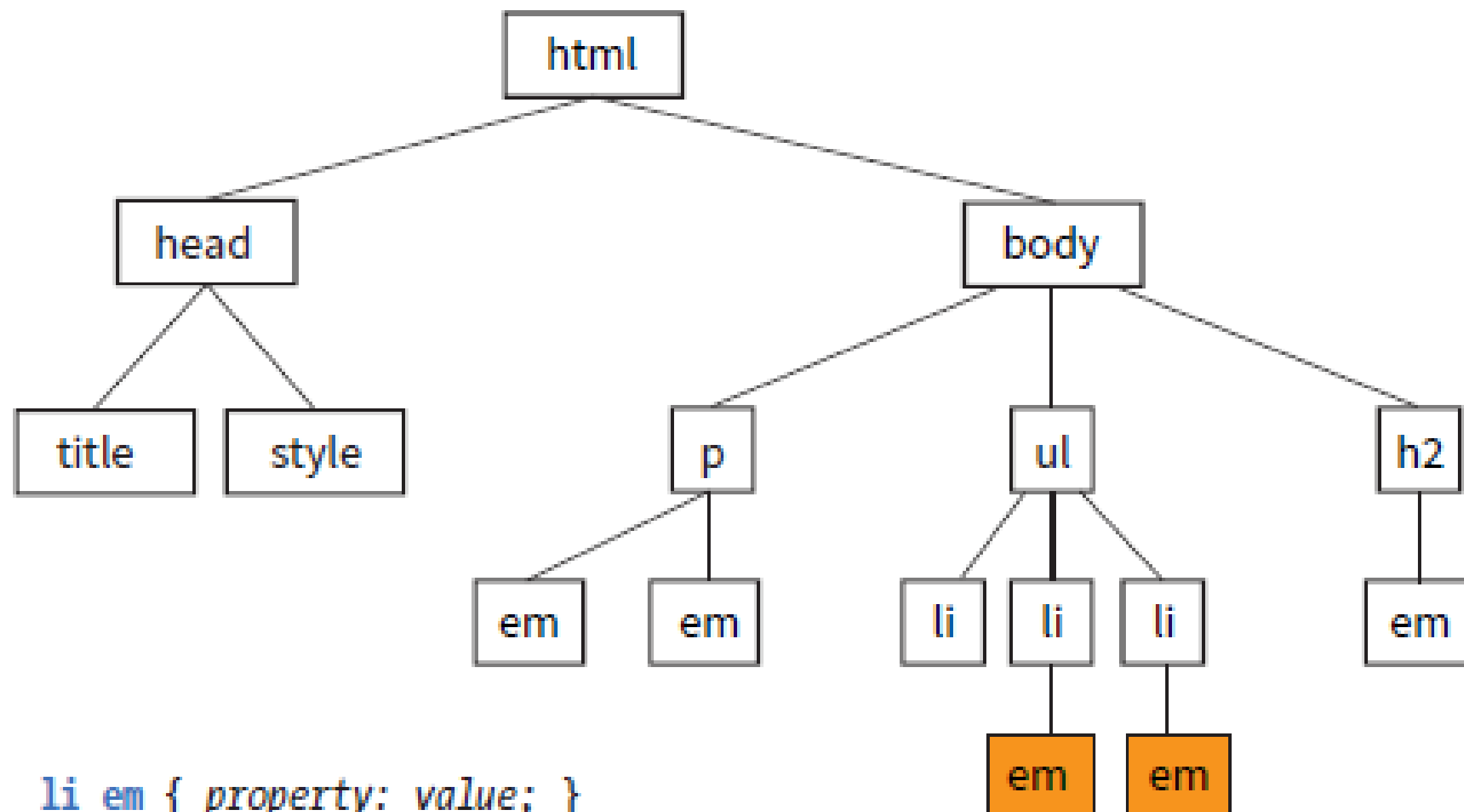


FIGURE 12-10. Only `em` elements within `li` elements are selected. The other `em` elements are unaffected.

Descendent Selectors (cont'd)

They can appear as part of a grouped selector:

```
h1 em, h2 em, h3 em {color: red;}
```

(only emphasized text in h1, h2, and h3 elements)

They can be several layers deep:

```
ol a em {font-variant: small-caps;}
```

(only emphasized text in links in ordered lists)

ID Selectors

ID selectors (indicated by a # symbol) target elements based on the value of their ID attributes:

```
<li id="primary">Primary color t-shirt</li>
```

To target just that item:

```
li#primary {color: olive;}
```

To omit the element name:

```
#primary {color: olive;}
```

It can be used as part of a compound or contextual selector:

```
#intro a { text-decoration: none;}
```

Class Selectors

Class selectors (indicated by a . symbol) select elements based on the value of their class attributes:

```
p.special { color: orange; }
```

(All paragraphs with the class name "special" would be orange.)

To target *all* element types that share a class name, omit the element name in the selector:

```
.highlight { background-color: yellow; }
```

(All elements with the class "highlight" would have a yellow background.)

Universal Selector

The **universal element selector** (*) matches any element, like a wildcard in programming languages:

```
* {border: 1px solid gray;}
```

(puts a 1-pixel gray border around every element in the document)

Can be used as part of contextual selectors:

```
#intro * {border: 1px solid gray;}
```

(selects all elements contained within an element with the ID **intro**)

Other Contextual Selectors

Descendant selectors are one of four types of contextual selectors (called **combinators** in the Selectors specifications Level 3 and Level 4). The other three are **child selectors**, **next-sibling selectors**, and **subsequent-sibling selectors**.

Child selector

A **child selector** is similar to a descendant selector, but it targets only the direct children of a given element. There may be no other hierarchical levels in between. They are indicated with the greater-than symbol (**>**). The following rule affects emphasized text, but only when it is directly contained in a **p** element. An **em** element inside a link (**a**) within the paragraph would not be affected.

```
p > em {font-weight: bold;}
```

Next-sibling selector

A **next-sibling selector** targets an element that comes directly after another element with the same parent. It is indicated with a plus (+) sign. This rule gives special treatment to paragraphs that follow an **h1**. Other paragraphs are unaffected.

```
h1 + p {font-style: italic;}
```

Subsequent-sibling selectors

A **subsequent-sibling selector** selects an element that shares a parent with the specified element and occurs after it in the source order. They do not need to follow one another directly. This type of selector is new in CSS3 and is not supported by Internet Explorer 8 and earlier. The following rule selects any **h2** that both shares a parent element (such as a **section** or **article**) with an **h1** and appears after it in the document.

```
h1 ~ h2 {font-weight: normal;}
```

Selectors Examples

- Adapted from https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Selectors and https://www.w3schools.com/css/css_selectors.asp

Selector Types

- Universal
- Type/Element
- ID
- Class
- Combinator
- Pseudo-class
- Pseudo-element
- Attribute

Universal

```
<!DOCTYPE html>
<html>
<head>
<style>
* {
  text-align: center;
  color: blue;
}
</style>
</head>
<body>
```

```
<h1>Hello world!</h1>
```

```
<p>Every element on the page will be affected by the style.</p>
```

```
<p id="para1">Me too!</p>
```

```
<p>And me!</p>
```

```
</body>
```

```
</html>
```

Hello world!

Every element on the page will be affected by the style.

Me too!

And me!

Element/Type

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
p {  
  background-color: yellow;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Welcome to My Homepage</h1>
```

```
<div>
```

```
  <p id="firstname">My name is Donald.</p>
```

```
  <p id="hometown">I live in Duckburg.</p>
```

```
</div>
```

```
<p>My best friend is Mickey.</p>
```

```
</body>
```

```
</html>
```

Welcome to My Homepage

My name is Donald.

I live in Duckburg.

My best friend is Mickey.

Grouped

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
h1, p {
```

```
  text-align: center;
```

```
  color: red;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Hello World!</h1>
```

```
<h2>Smaller heading!</h2>
```

```
<p>This is a paragraph.</p>
```

```
</body>
```

```
</html>
```

Hello World!

Smaller heading!

This is a paragraph.

(All types can be grouped)

ID

```
<!DOCTYPE html>
<html>
<head>
<style>
#firstname {
  background-color: yellow;
}
</style>
</head>
<body>
```

```
<h1>Welcome to My Homepage</h1>
```

```
<div class="intro">
  <p id="firstname">My name is Donald.</p>
  <p id="hometown">I live in Duckburg.</p>
</div>
```

```
<p>My best friend is Mickey.</p>
```

```
</body>
</html>
```

Welcome to My Homepage

My name is Donald.

I live in Duckburg.

My best friend is Mickey.

Class

```
<!DOCTYPE html>
<html>
<head>
<style>
.intro {
  background-color: yellow;
}
</style>
</head>
<body>

<h1>Welcome to My Homepage</h1>

<div class="intro">
  <p>My name is Donald.</p>
  <p>I live in Duckburg.</p>
</div>

<p>My best friend is Mickey.</p>

<p class="intro">My best friend is Mickey.</p>

</body>
</html>
```

Welcome to My Homepage

My name is Donald.

I live in Duckburg.

My best friend is Mickey.

My best friend is Mickey.

Class Specific to Element

```
<!DOCTYPE html>
<html>
<head>
<style>
p.intro {
  background-color: yellow;
}
</style>
</head>
<body>

<h1>Welcome to My Homepage</h1>

<div class="intro">
  <p>My name is Donald.</p>
  <p>I live in Duckburg.</p>
</div>

<p>My best friend is Mickey.</p>

<p class="intro">My best friend is Mickey.</p>

</body>
</html>
```

Welcome to My Homepage

My name is Donald.

I live in Duckburg.

My best friend is Mickey.

My best friend is Mickey.

Class (Multiple)

This heading will not be affected

This paragraph will be red and center-aligned.

This paragraph will be red,
center-aligned, and in a large
font-size.

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}

p.large {
  font-size: 300%;
}
</style>
</head>
<body>

<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned.</p>
<p class="center large">This paragraph will be red, center-aligned, and in a
large font-size.</p>

</body>
</html>
```

Contextual

Child $li > a \quad \{ \}$

Descendent $p \ a \quad \{ \}$

Adjacent sibling $h1 + p \quad \{ \}$

General $h1 \sim p \quad \{ \}$

END OF LECTURE